

7.2 Convergence of Series

A series converges absolutely when the sum of the moduli of its terms converges. The ratio test gives a sufficient condition.

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$$

When L exceeds one the series diverges, and when L equals one the test is inconclusive. The root test uses the n -th root instead.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

which follows from the Euler series by splitting even and odd terms.